WORKED SOLUTION

STEP I 1995, Question 12

Probability and Statistics • Arrangements and Counting

STEP I 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q12

The key idea

Every arrangement of the n pupils is equally likely, so each probability is a count of favourable arrangements divided by $n!$. Better still, by symmetry we only need to care about *which positions* the hockey players occupy: each of the $\binom{n}{r}$ sets of r positions is equally likely. Each part then becomes a counting problem about choosing positions. For the last part, the hint about gaps is the standard "stars and bars" trick: line up the non-hockey-players first, and the hockey players must go into separate gaps between them.

Solution

Throughout, call the r hockey players H and the other $n-r$ pupils N. All $n!$ orderings of the (distinguishable) pupils are equally likely. Since every set of r positions for the Hs arises from the same number, $r!\,(n-r)!$, of orderings, the set of positions occupied by the Hs is equally likely to be any of the $\binom{n}{r}$ possible sets. We use whichever view is more convenient.

Part (i)

Fill the two end positions first. The pupil at the left end is a hockey player with probability $\frac{r}{n}$. Given that, the pupil at the right end is one of the remaining $n-1$ pupils, each equally likely, of whom $r-1$ play hockey. So

$$\mathrm{P}(\text{hockey player at each end}) = \frac{r}{n}\cdot\frac{r-1}{n-1} = \frac{r(r-1)}{n(n-1)}.$$

(We need $r \geq 2$ for this to be non-zero, which the question guarantees.)

Part (ii)

The hockey players stand together exactly when their positions form a block of r consecutive places. Such a block can start at position $1, 2, \ldots, n-r+1$, so there are $n-r+1$ favourable sets of positions out of $\binom{n}{r}$:

$$\mathrm{P}(\text{all together}) = \frac{n-r+1}{\binom{n}{r}} = \frac{r!\,(n-r+1)!}{n!}.$$

Equivalently, glue the r hockey players into one "super-pupil": there are $(n-r+1)!$ orderings of the $n-r+1$ units and $r!$ orderings inside the block, giving the same answer. As a check, when $r = n$ this is 1, as it should be.

Part (iii)

Stand the $n-r$ non-hockey-players in a row first. They create $n-r+1$ gaps: one before the first, one between each neighbouring pair, and one after the last.



No two hockey players are adjacent if and only if each gap receives at most one hockey player. (If two Hs were in the same gap they would be next to each other; if they are in different gaps, at least one N separates them.) So the favourable arrangements correspond to choosing r of the $n-r+1$ gaps, one H per chosen gap.

Count positions: a favourable set of H-positions is determined by which r gaps are used, so there are $\binom{n-r+1}{r}$ of them. Hence

$$\mathrm{P}(\text{no two together}) = \frac{\binom{n-r+1}{r}}{\binom{n}{r}} = \frac{(n-r+1)!\,(n-r)!}{(n-2r+1)!\,n!}.$$

(Counting orderings of the pupils directly gives the same thing: $(n-r)!$ ways to order the Ns, then $(n-r+1)(n-r)\cdots(n-2r+2) = \frac{(n-r+1)!}{(n-2r+1)!}$ ways to put the r distinct Hs into distinct gaps, all over $n!$.)

Ask yourself When is it impossible to keep all the hockey players apart?

We need r different gaps, and there are only $n-r+1$ of them. So:

- if $r \le n-r+1$, that is $n \ge 2r-1$, the probability is the non-zero value above;
- if $r > n-r+1$, that is $n \le 2r-2$ (too few non-hockey-players to separate them), the probability is 0.

The boundary case $n = 2r-1$ forces the pattern $HNHN\cdots NH$, and the formula gives $\frac{r!\,(r-1)!}{(2r-1)!} = 1/\binom{2r-1}{r}$, the chance that the Hs land in exactly those r alternate places.

Checks

- Take $n = 10$, $r = 3$. Then (i) gives $\frac{6}{90} = \frac{1}{15}$; (ii) gives $\frac{8}{120} = \frac{1}{15}$; (iii) gives $\binom{8}{3}/\binom{10}{3} = \frac{56}{120} = \frac{7}{15}$. All three agree with a computer enumeration of the 120 possible sets of positions, and the formulae were checked this way for every $2 \leq r \leq n \leq 14$ (and against all $n!$ orderings for $n \leq 7$).
- With $r = 2$, parts (ii) and (iii) are complementary events (the two players are either adjacent or not), and indeed $\frac{2}{n} + \frac{(n-1)(n-2)}{n(n-1)} = \frac{2}{n} + \frac{n-2}{n} = 1$.

Common slips

- Counting only $n - r - 1$ gaps (the ones strictly between non-hockey-players) and forgetting the two ends of the row.
- Mixing models: counting H-positions in the numerator but full orderings of pupils ($n!$) in the denominator. Choose one view and keep to it.
- Giving the condition for a non-zero answer as $n \geq 2r$; the case $n = 2r - 1$ (alternating, starting and ending with H) also works.

Beyond STEP

The gap argument is the "stars and bars" idea, a workhorse of enumerative combinatorics; the same count $\binom{n-r+1}{r}$ answers "how many r-subsets of $\{1, \ldots, n\}$ contain no two consecutive integers", and summing it over r gives the Fibonacci numbers. The step of replacing orderings of people by sets of positions is a first taste of exploiting symmetry to shrink a sample space, used constantly in university probability. Blitzstein and Hwang, *Introduction to Probability*, has excellent chapters on counting and the naive definition of probability.

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